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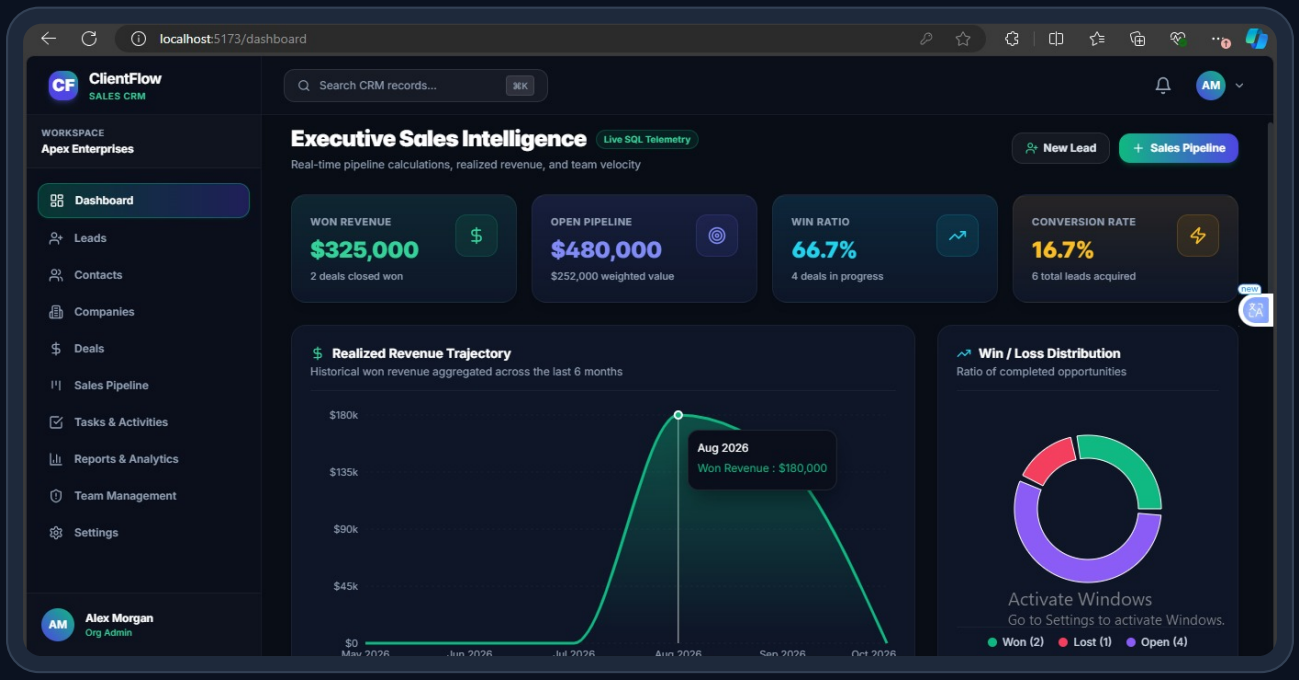
ClientFlow

ENTERPRISE CRM

System Documentation

Enterprise CRM & Sales Management System

Technical Architecture, Feature Walkthrough & Operational Manual



React 18 · TypeScript · Node.js · Express · PostgreSQL · Prisma · Tailwind CSS

VERSION 1.0.0 · FULL-STACK MONOREPO SPECIFICATION · OCTOBER 2026

Turn relationships into revenue.

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About this document

This manual documents **ClientFlow**, a multi-tenant B2B CRM and sales management platform. Every screenshot was captured from the running application against the seeded **Apex Enterprises** demo organization, so the numbers you see in the figures (won revenue, pipeline value, win rate) are real, database-calculated values rather than mock-ups.

1 Executive Overview

One workspace for the entire revenue cycle, from first contact to closed deal.

Small and medium-sized enterprises frequently struggle with fragmented customer data spread across spreadsheets, disconnected communication channels and opaque sales projections. **ClientFlow** is a complete, scalable and secure SaaS solution that streamlines the whole revenue cycle. From the moment a prospect reaches out to deal closure and ongoing customer success, it brings leads, deals, contacts, pipeline stages, tasks and analytics into a single high-performance workspace.

4	18	17/17	3
Role-based access levels	Documented REST endpoints	Integration tests passing	Monorepo workspaces

Core value propositions

Zero data leakage

Strict multi-tenant boundaries. Every query is scoped by the organization id carried in the verified JWT, so one tenant can never read another tenant's records.

Qualification-to-pipeline velocity

One-click atomic conversion turns a qualified lead into a Contact, a Company and a Deal in a single all-or-nothing transaction.

Visual sales acceleration

An interactive drag-and-drop Kanban pipeline with optimistic updates and live revenue, deal-count and win-probability calculations per stage.

Executive Obsidian aesthetic

A modern B2B SaaS interface with an obsidian dark theme, high-contrast typography and jewel-toned KPI widgets built for long working sessions.

Key business objectives

- **Lead management and qualification:** capture incoming leads, prioritise prospects, detect duplicate records and convert qualified leads into customer accounts with a single click.
- **Visual sales pipeline:** manage opportunity stages on an optimistic-update Kanban board.
- **Account governance:** maintain structured directories of customer Contacts and target Company accounts.
- **Team collaboration:** assign tasks, track follow-up deadlines and enforce multi-tenant Role-Based Access Control (RBAC).
- **Executive intelligence:** live database metrics, win-rate tracking, weighted revenue forecasting and sanitised CSV exports.

2 Architecture & Technology Stack

A production-minded, type-safe stack designed for performance, scaling and operational predictability.

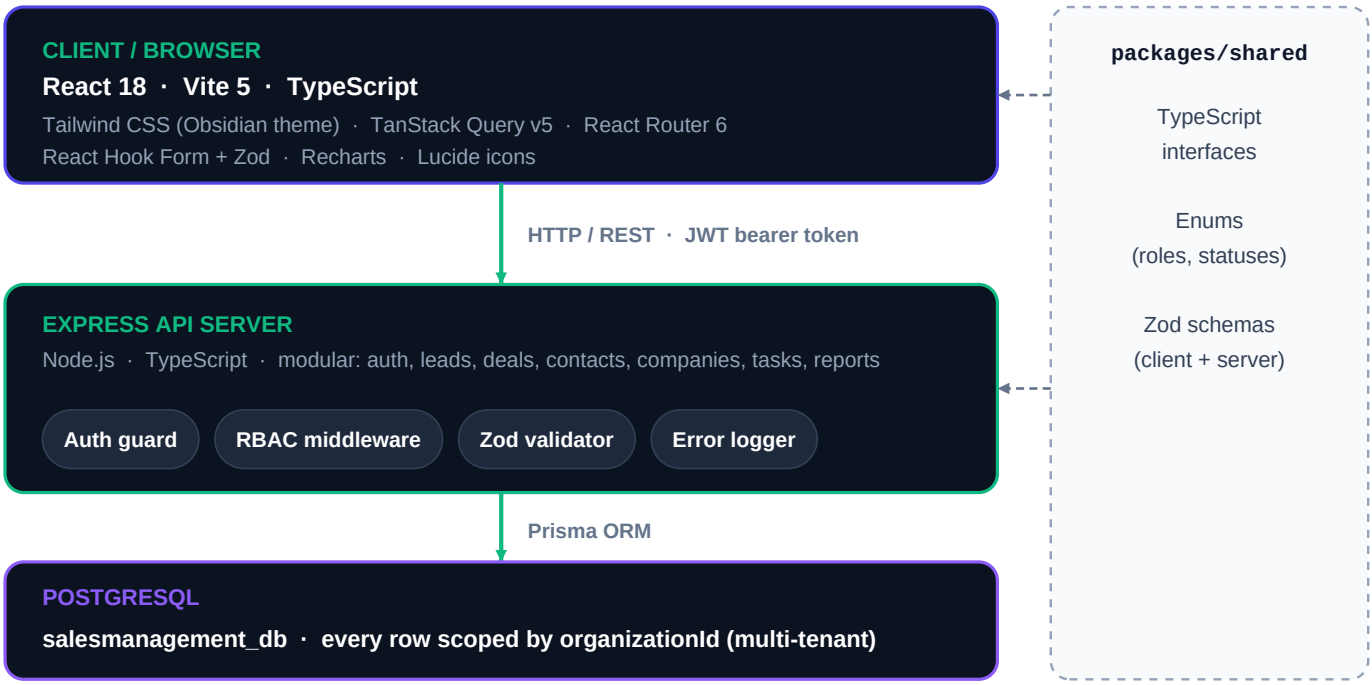


Figure 2.1 — System architecture: browser, Express API, PostgreSQL and the shared type package

Technology stack

Layer	Technology selected	Purpose
Frontend	React 18 + TypeScript + Vite 5	High-performance single-page interface
Styling	Tailwind CSS (Obsidian Executive theme)	Custom B2B SaaS dark-theme aesthetic
State & data	TanStack React Query v5	Server-state caching, background revalidation, optimistic mutations
Forms	React Hook Form + Zod	Typed form handling and schema validation
Visuals	Recharts + Lucide React	Charts, KPI widgets and iconography
Backend API	Node.js + Express.js (TypeScript)	RESTful API layer with Zod request validation
Database & ORM	PostgreSQL + Prisma ORM	Relational persistence with multi-tenant scoping
Security	JWT + Argon2id / bcrypt, Helmet, CORS, rate limiting	Secure tokens, password hashing and abuse protection
Testing	Jest + Supertest	Automated API integration test suite

Request lifecycle

Every authenticated API call passes through the same four guards before touching data. The final step always applies the tenant filter, which is what makes cross-tenant leakage structurally impossible rather than a matter of developer discipline.

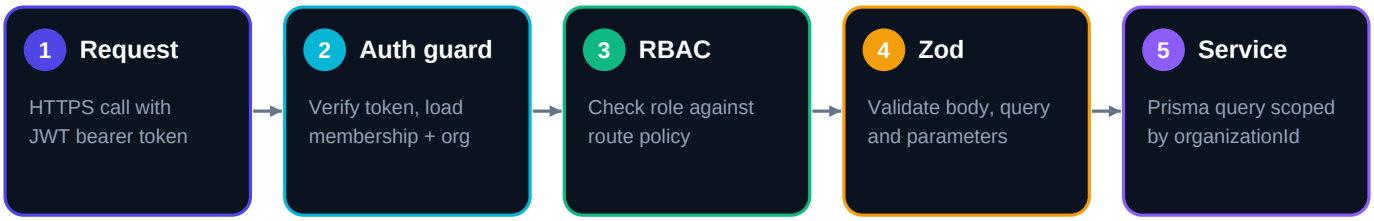


Figure 2.2 — Middleware pipeline applied to every protected endpoint

Project structure

```
clientflow/
├── apps/
│   ├── web/                    # React 18 + Vite single-page application
│   │   ├── public/doc.html     # Printable documentation (PDF export source)
│   │   └── src/
│   │       ├── api/            # Axios client with automatic token refresh
│   │       ├── components/     # Button, Modal, Card, Badge ...
│   │       ├── context/        # AuthContext, ToastContext
│   │       ├── layouts/        # AppShell (sidebar, topbar, global search)
│   │       ├── pages/          # Landing, Login, Dashboard, Leads, Deals, Pipeline ...
│   │       └── routes/         # Protected and public routes
│   └── api/                    # Node.js + Express REST API
│       ├── src/
│       │   ├── config/         # Environment, logger, Prisma client
│       │   ├── middleware/     # Auth, RBAC, validation, errors, rate limiting
│       │   ├── modules/        # Auth, Leads, Deals, Contacts, Companies, Tasks, Reports
│       │   ├── tests/          # Supertest API suite
│       │   └── app.ts          # Express application configuration
│       └── prisma/
│           ├── schema.prisma   # Multi-tenant PostgreSQL schema
│           └── seed.ts         # Demo B2B data seeding script
├── packages/shared/            # Shared types, Zod schemas and enums
├── docker-compose.yml          # Containerised PostgreSQL service
├── package.json                # Monorepo workspaces configuration
└── .env                        # Environment configuration
```

3 Data Model & Multi-Tenant Scoping

Every entity belongs to an Organization, and the backend enforces that boundary automatically.

The backend middleware extracts the tenant id from the verified JWT payload and injects it into every Prisma query. Controllers never accept an organization id from the request body, so a client cannot ask for data outside its own workspace.

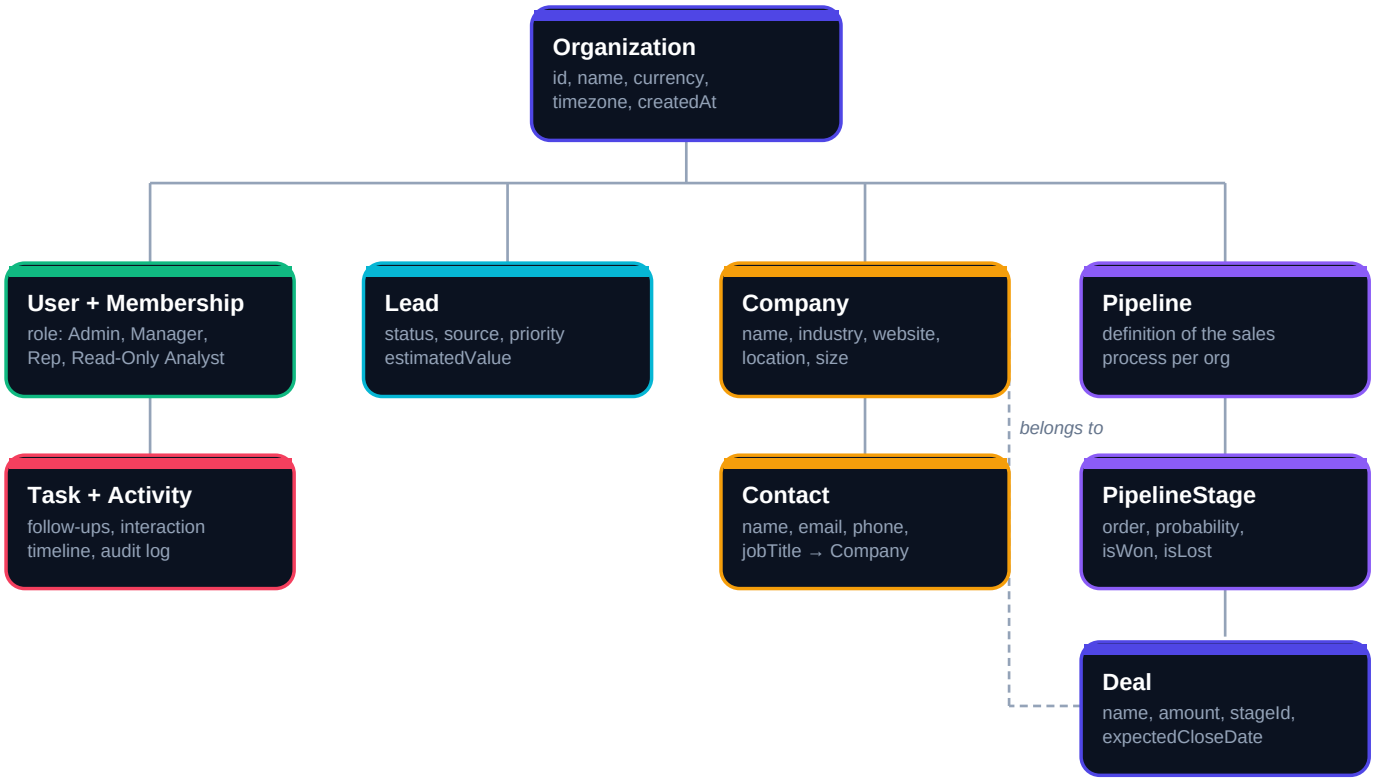


Figure 3.1 — Core entities and their relationships (all rows carry organizationId)

Entity	Purpose	Key fields
Organization	Workspace tenant root	id, name, currency, timezone, createdAt
User & Membership	Identities and role permissions	ORG_ADMIN, SALES_MANAGER, SALES_REP, READ_ONLY_ANALYST
Lead	Prospective customers	firstName, lastName, companyName, status, source, priority, estimatedValue
Company	Target accounts	name, industry, website, location, size
Contact	Customer stakeholders linked to a Company	firstName, lastName, email, phone, jobTitle
Pipeline & PipelineStage	Sales process definition	order, probability, isWon, isLost
Deal	Revenue opportunities	name, amount, stageId, expectedCloseDate
Task & Activity	Scheduled follow-ups and interaction timeline	type, priority, dueDate, completion state, audit entries

4 Feature Walkthrough

A guided tour of every screen, captured from the live application.

4.1 Public Landing Page & Role Sandbox

The marketing shell introduces the product with a high-impact hero, a live preview of the CRM and direct entry points. The top navigation offers **Sign In**, **Register** and a **Launch Live System** shortcut, while the hero presents the two primary calls to action, *Get Started Free* and *Launch Live System*.

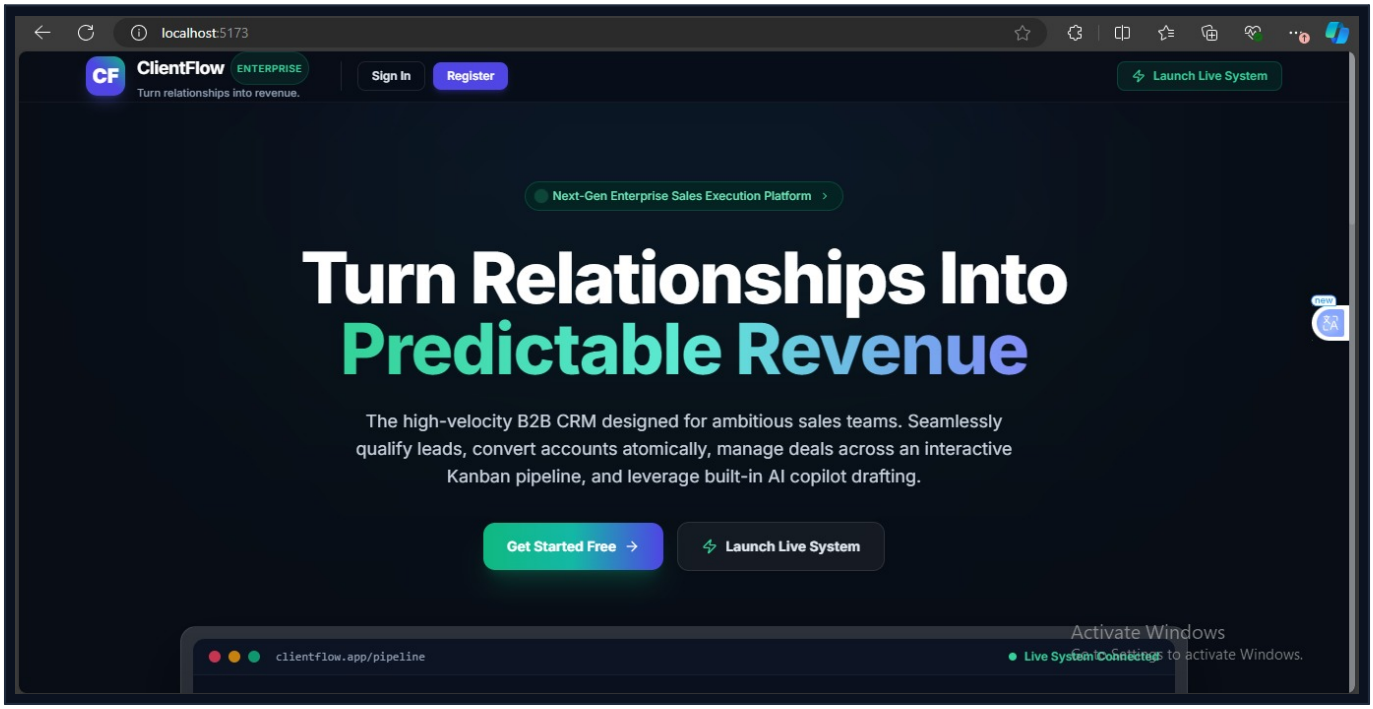


Figure 4.1 — Landing page hero: value proposition, enterprise badge and primary calls to action

Beneath the hero, an interactive CRM preview card mirrors the real pipeline view: KPI tiles for won revenue and open pipeline, mini Kanban columns and an AI sales insight banner that nudges reps about leads awaiting follow-up.

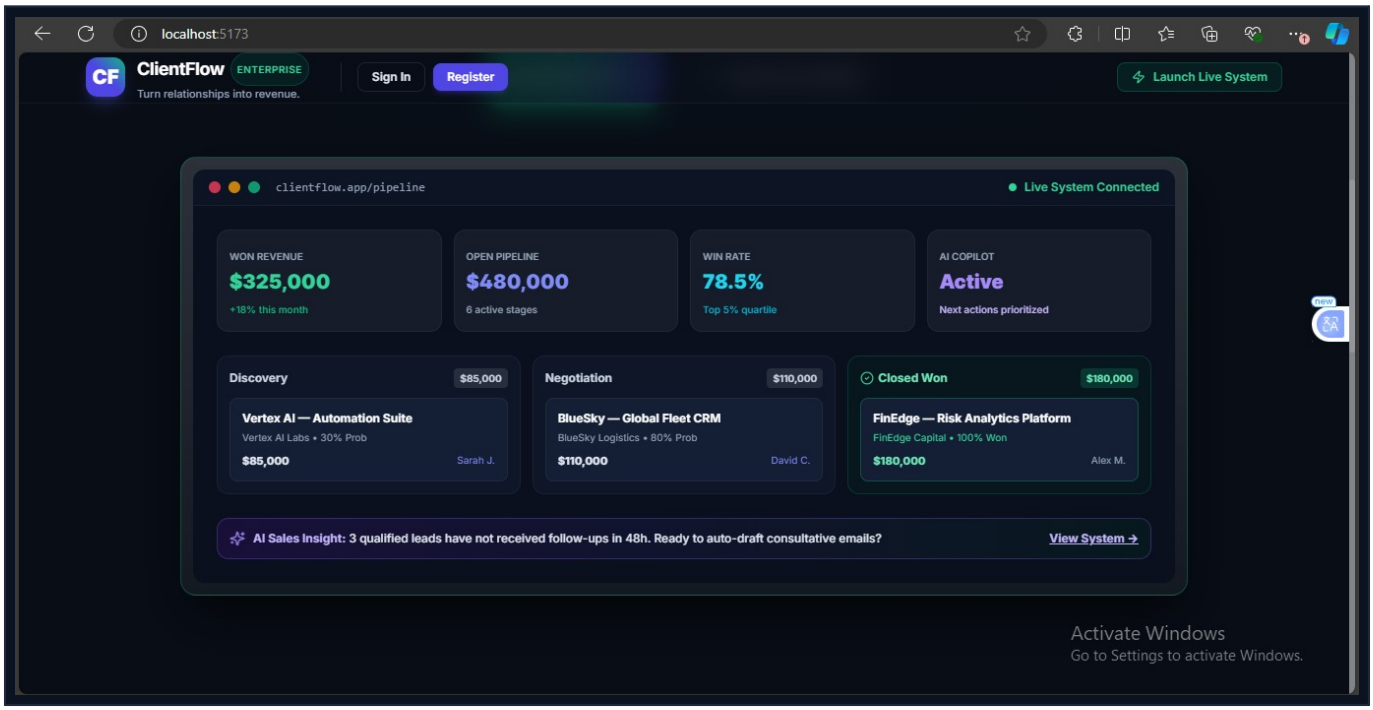


Figure 4.2 — Live CRM preview card with KPI tiles, pipeline columns and AI Sales Insight

The **Enterprise Role-Based Access Control** section doubles as a sandbox: each role card links straight to the sign-in portal so reviewers can explore the product from the perspective of an Admin, Manager, Representative or Analyst.

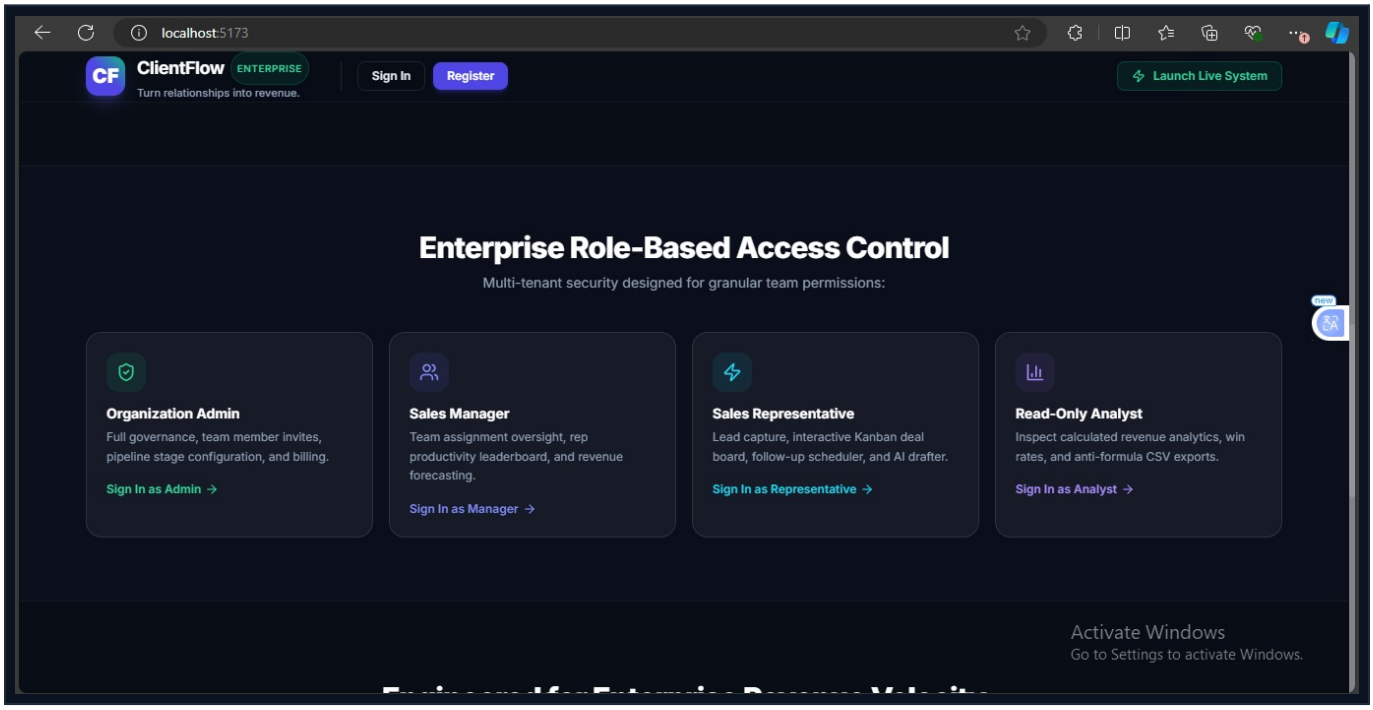


Figure 4.3 — Role sandbox cards: Organization Admin, Sales Manager, Sales Representative and Read-Only Analyst

The feature grid summarises the engineering principles behind the product: an interactive Kanban board with optimistic rendering and backend rollback, atomic lead conversion, an AI sales assistant for tone-tailored follow-up emails, real SQL-calculated analytics, strict tenant isolation enforced through JWT middleware, and follow-up lists with overdue alerts.

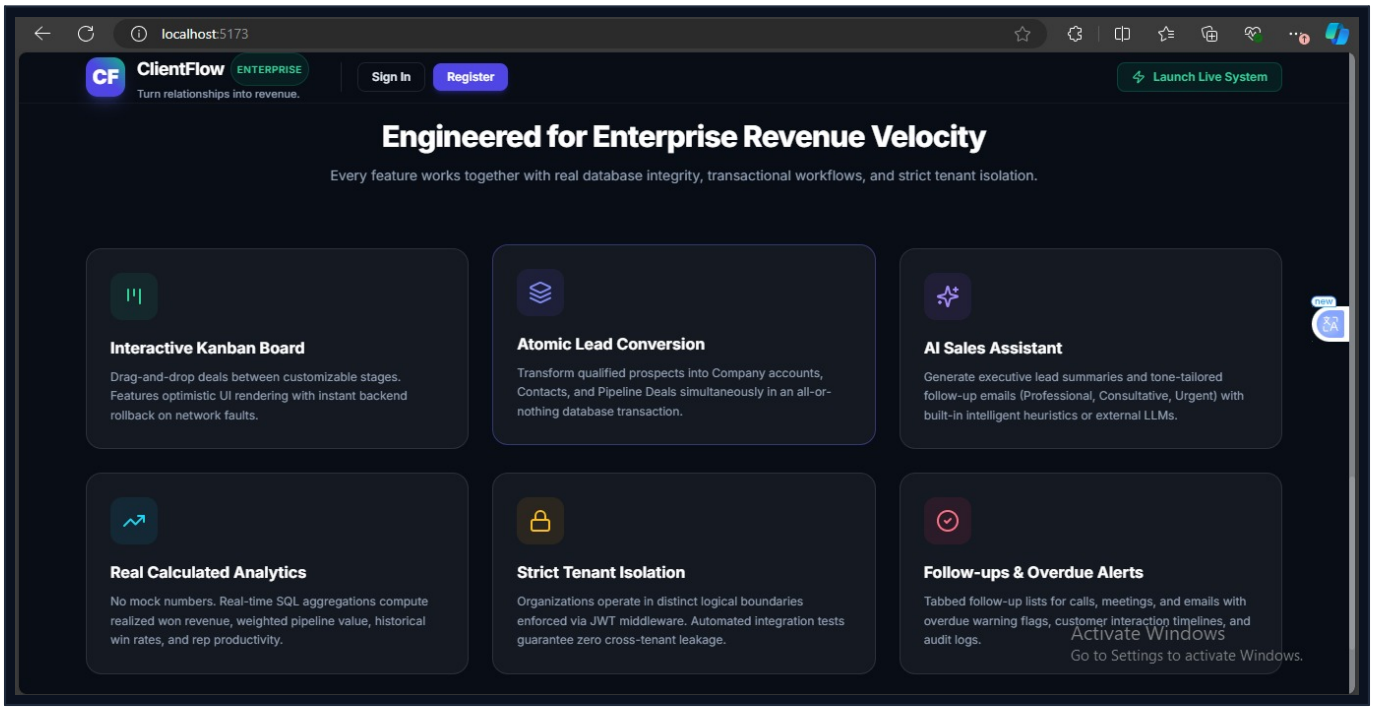


Figure 4.4 — "Engineered for Enterprise Revenue Velocity": the six headline capabilities

4.2 Authentication & Onboarding

Access is gated by JWT-based authentication. Returning users sign in with a work email and password (with a show/hide toggle and a forgot-password link), while new customers create a fully isolated organization from the registration page.

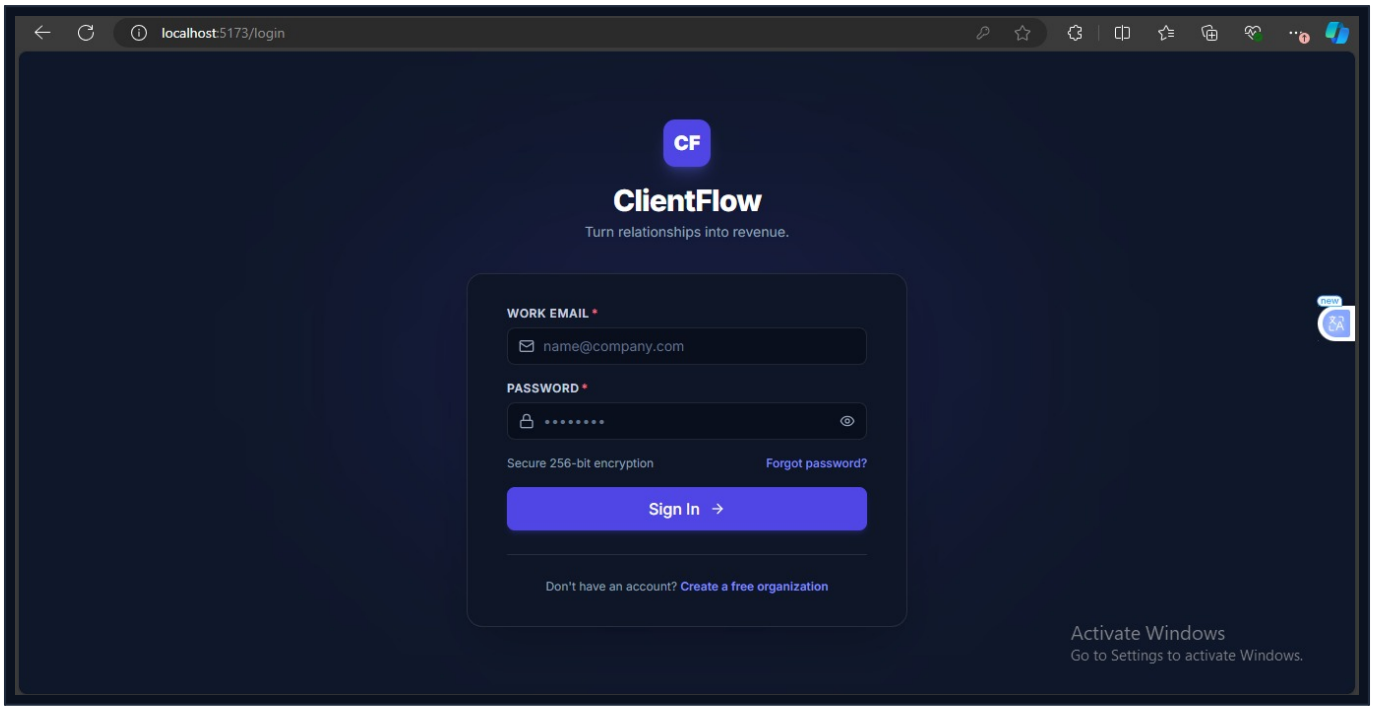


Figure 4.5 — Sign-in screen with work email, password and security badge

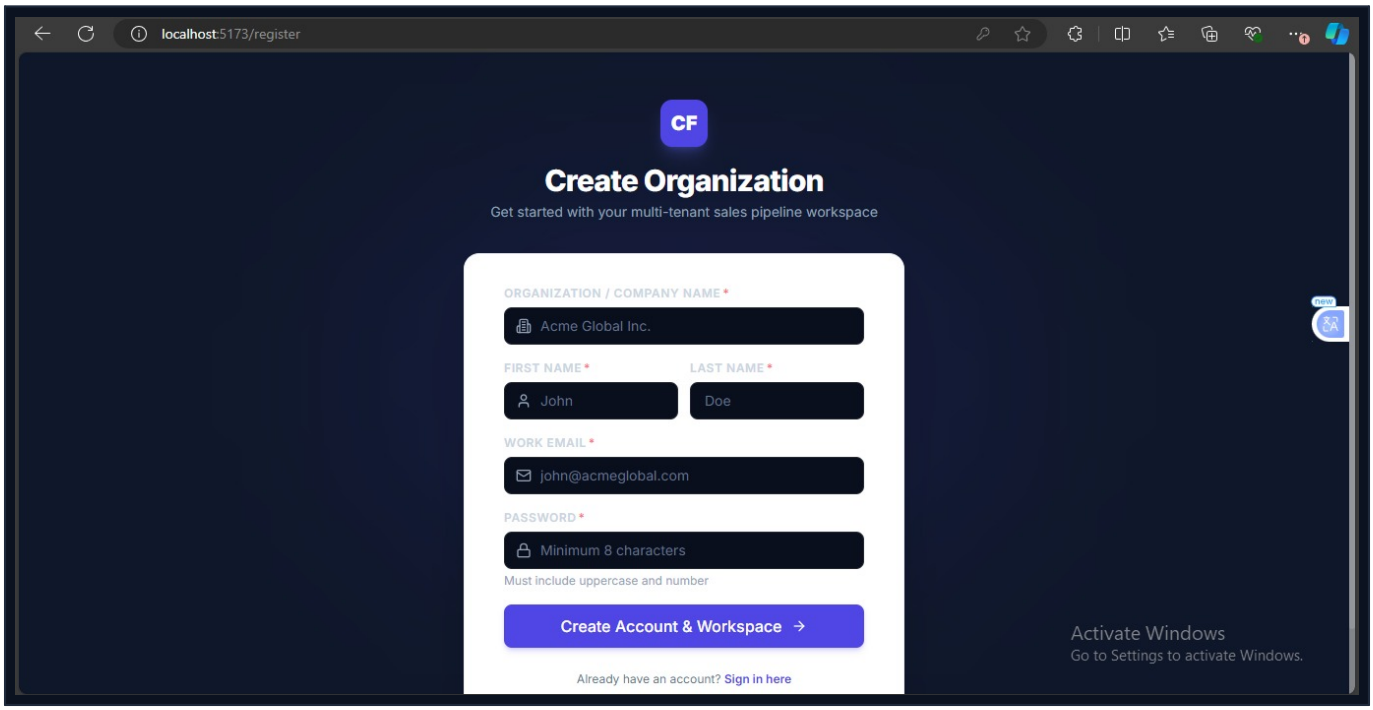


Figure 4.6 — Create Organization: company name, administrator details and password policy (minimum 8 characters, one uppercase letter and one number)

What happens on registration

Submitting the form creates the **Organization** (the new tenant), the first **User** with the **Organization Admin** role and a default sales pipeline, then signs the administrator in. Passwords are hashed with Argon2id / bcrypt (12 salt rounds) and are never stored or logged in plain text.

4.3 Executive Dashboard

The dashboard is the command centre of the workspace. Its **Live SQL Telemetry** badge signals that every figure is aggregated from the database on request, not hard-coded. Global search (⌘K) and the New Lead and Sales Pipeline shortcuts are available from every screen.

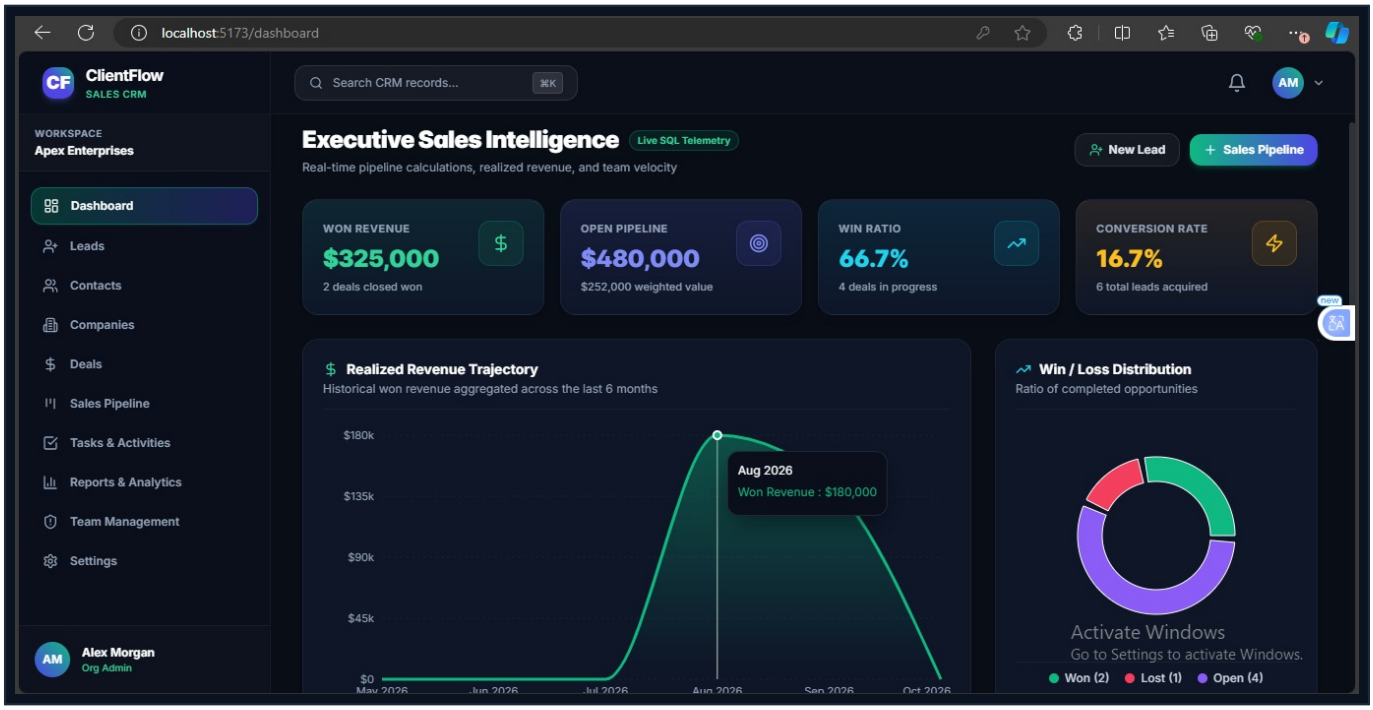


Figure 4.7 — Executive Sales Intelligence dashboard for the Apex Enterprises workspace (Org Admin view)

KPI	Value in demo data	How it is calculated
Won revenue	$325,000	Sum of deals in a Closed Won stage (2 deals)
Open pipeline	$480,000	Sum of deals in open stages; $252,000 probability-weighted
Win ratio	66.7%	Won ÷ (Won + Lost) completed opportunities, with 4 deals still in progress
Conversion rate	16.7%	Converted leads ÷ total leads acquired (1 of 6)

- **Realized Revenue Trajectory:** an area chart of won revenue aggregated across the last six months, with hover tooltips.
- **Win / Loss Distribution:** a donut chart splitting completed and open opportunities (Won 2, Lost 1, Open 4).
- **Further widgets:** scrolling down reveals pipeline value by stage, a sales leaderboard, high-priority tasks and a live activity stream (below).

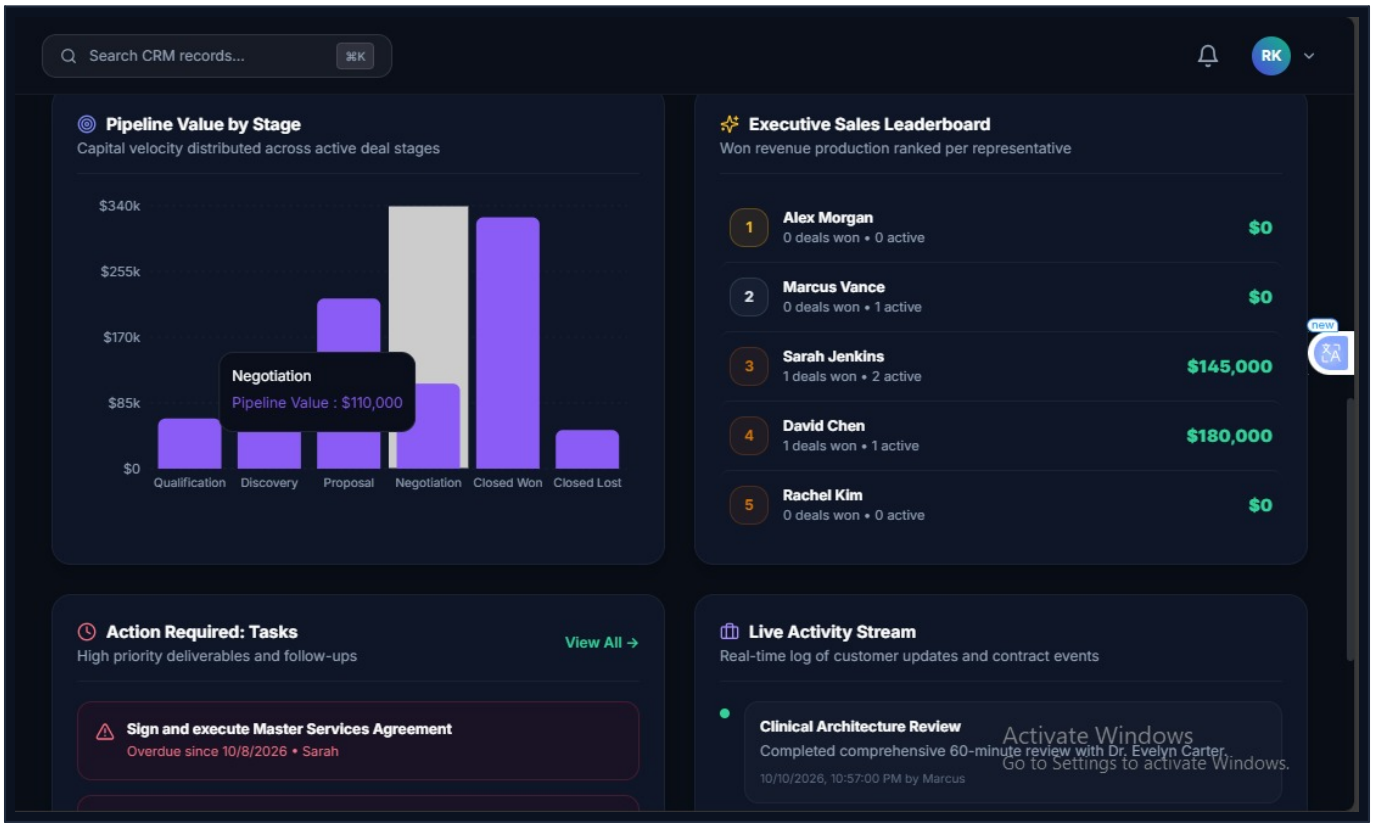


Figure 4.7b — Lower dashboard: Pipeline Value by Stage, Executive Sales Leaderboard, Action Required tasks and Live Activity Stream

4.4 Lead Management & Atomic Conversion

Leads are raw prospects moving through qualification. The directory supports free-text search plus filters for **Status** (NEW, CONTACTED, QUALIFIED, UNQUALIFIED, CONVERTED), **Source** and **Priority**. Colour-coded badges make the state of each prospect readable at a glance, and a duplicate-email banner warns reps when several leads share the same address.

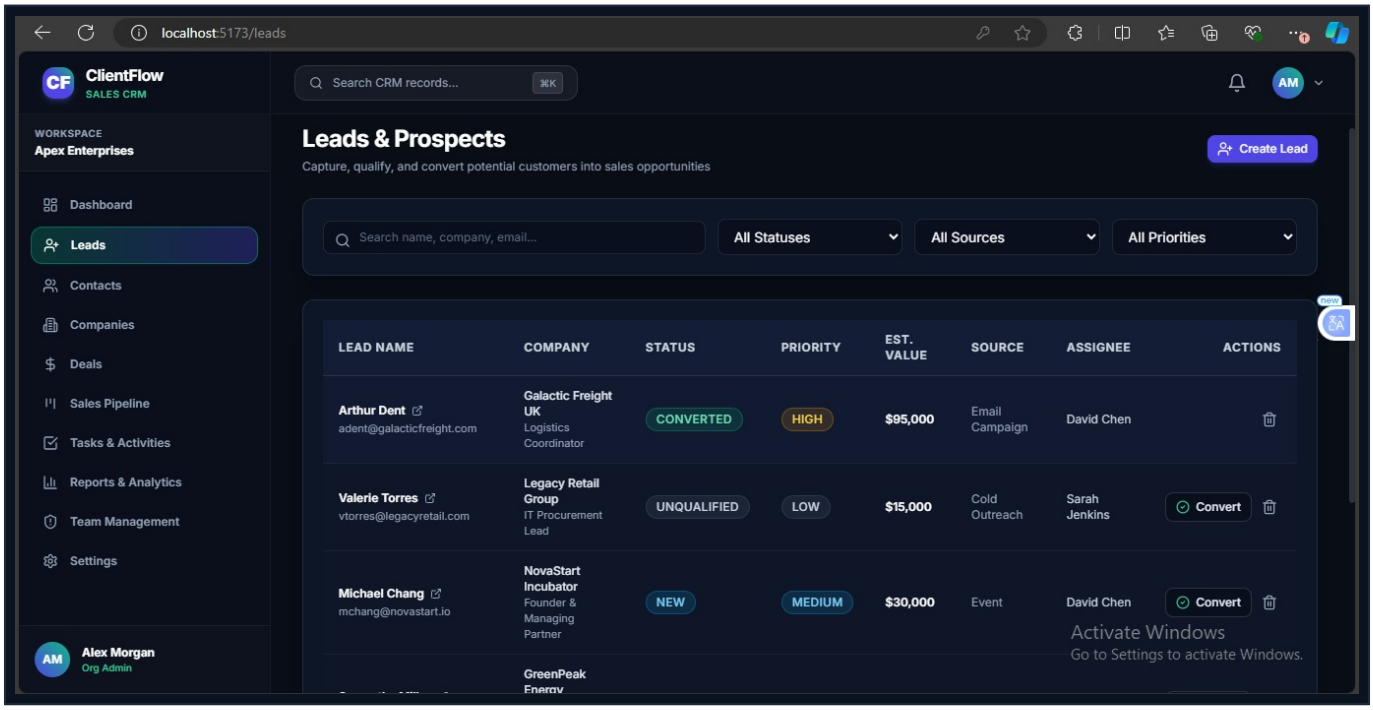


Figure 4.8 — Leads & Prospects: filters, status and priority badges, estimated value, source, assignee and the Convert action

Pressing **Convert** runs a single database transaction. Either every record is created and the lead is marked *CONVERTED*, or nothing is written at all, so the CRM can never contain a half-converted lead.

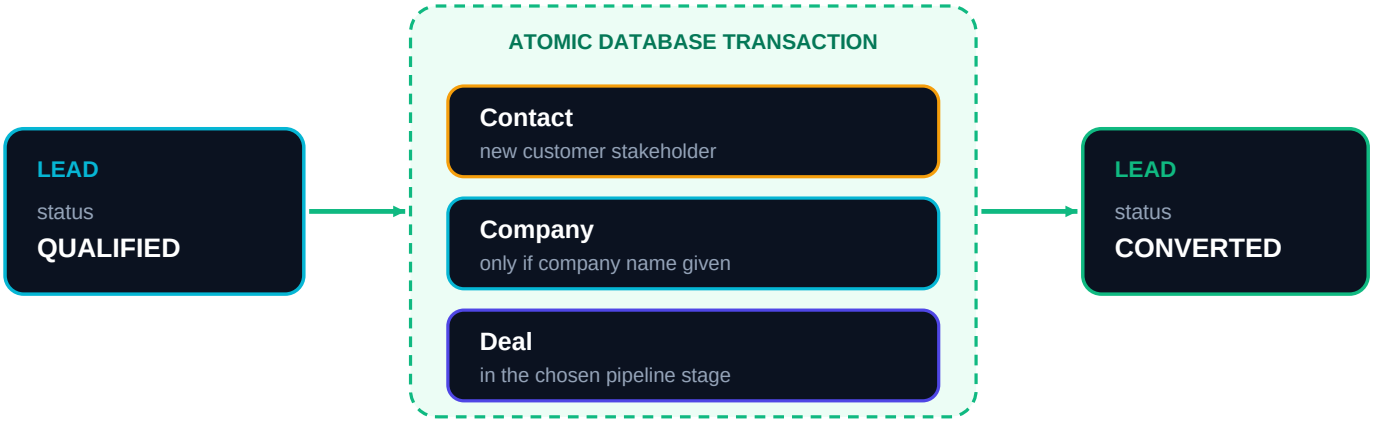


Figure 4.9 — Atomic lead conversion: all-or-nothing creation of Contact, Company and Deal

4.5 Contacts & Company Accounts

The **Contacts** directory lists verified decision-makers with their job title, linked company, email, phone number and account owner. The **Companies** view shows target accounts with industry, location and headcount band, plus live counts of linked contacts and deals.

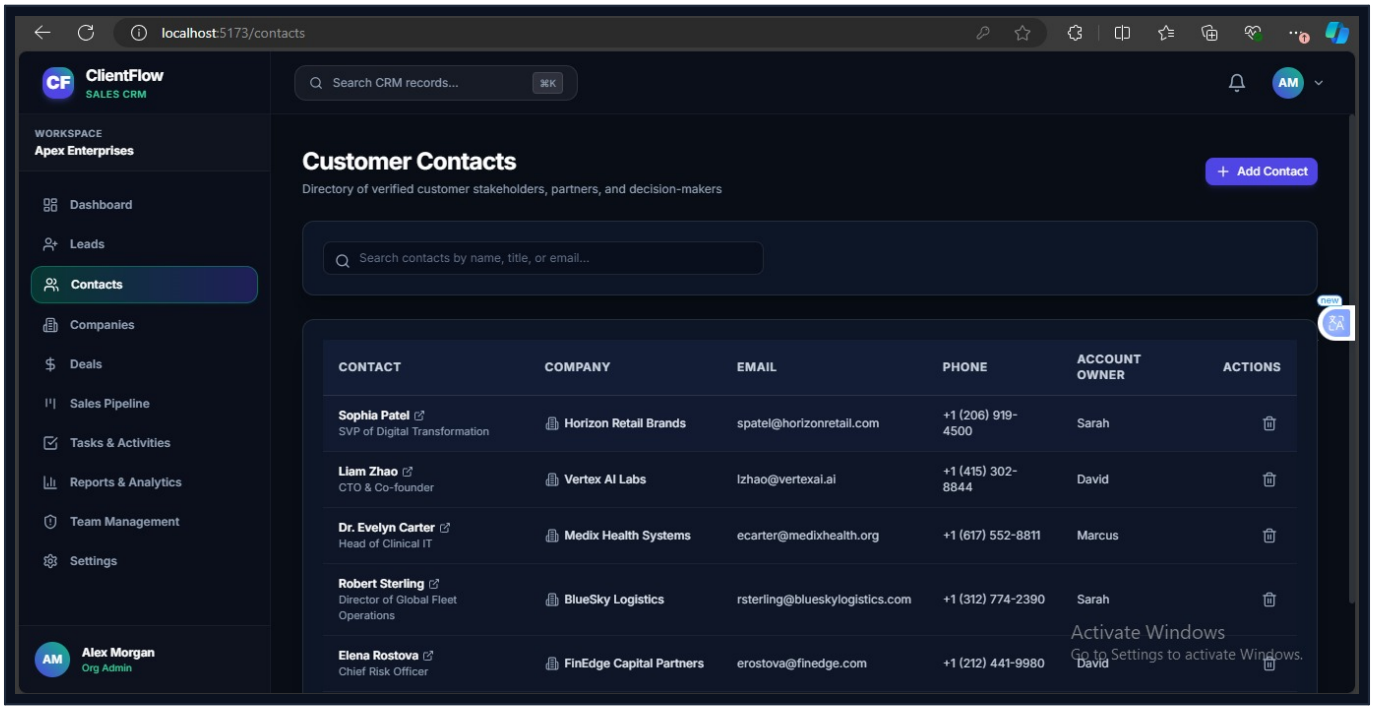


Figure 4.10 — Customer Contacts: stakeholders, titles, linked companies and account owners

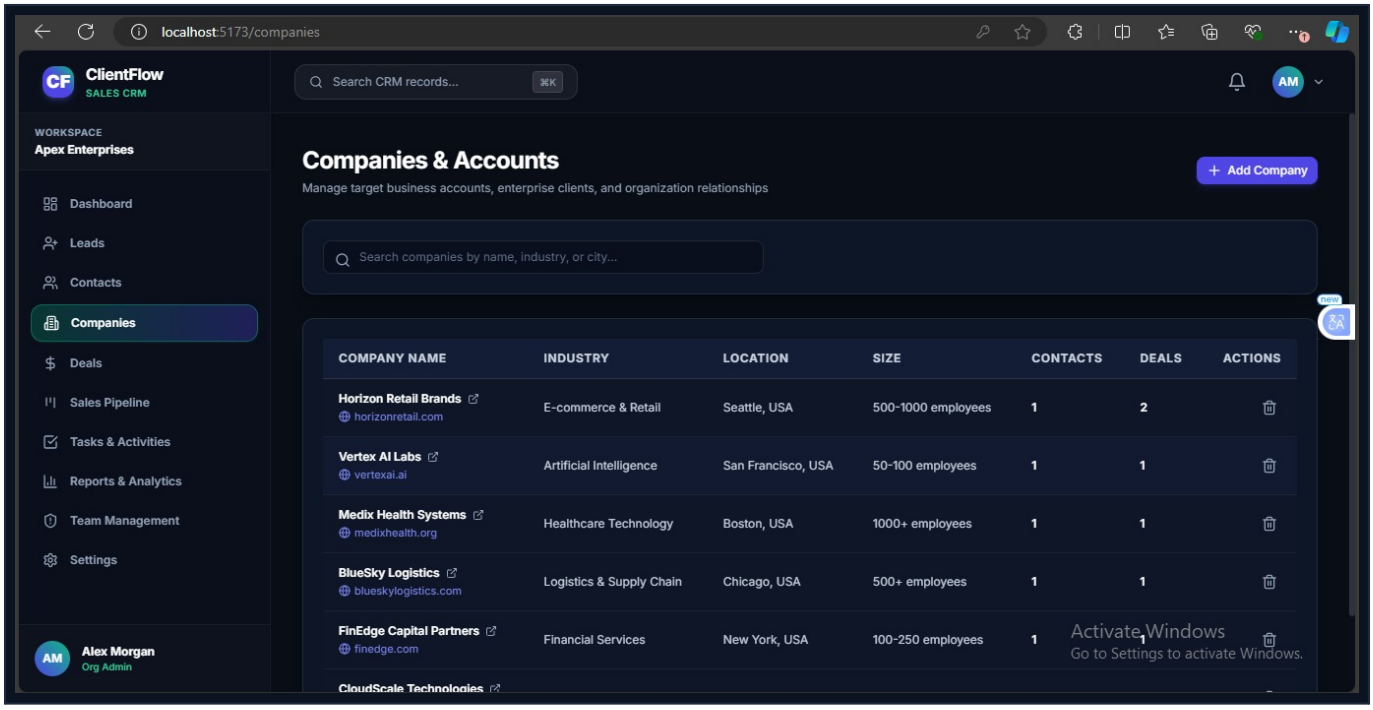


Figure 4.11 — Companies & Accounts: industry, location, size, contact count and active-deal count

4.6 Sales Pipeline (Kanban) & Deals

Opportunities move across the stages *Qualification, Discovery, Proposal, Negotiation, Closed Won* and *Closed Lost* by drag and drop. The UI updates **optimistically** the instant a card is dropped, while a `PATCH /deals/:id/stage` request synchronises the database; if the request fails, the card snaps back to its original column. Each column header shows the stage value, deal count and win probability, and the **Active Value** badge totals the open pipeline.

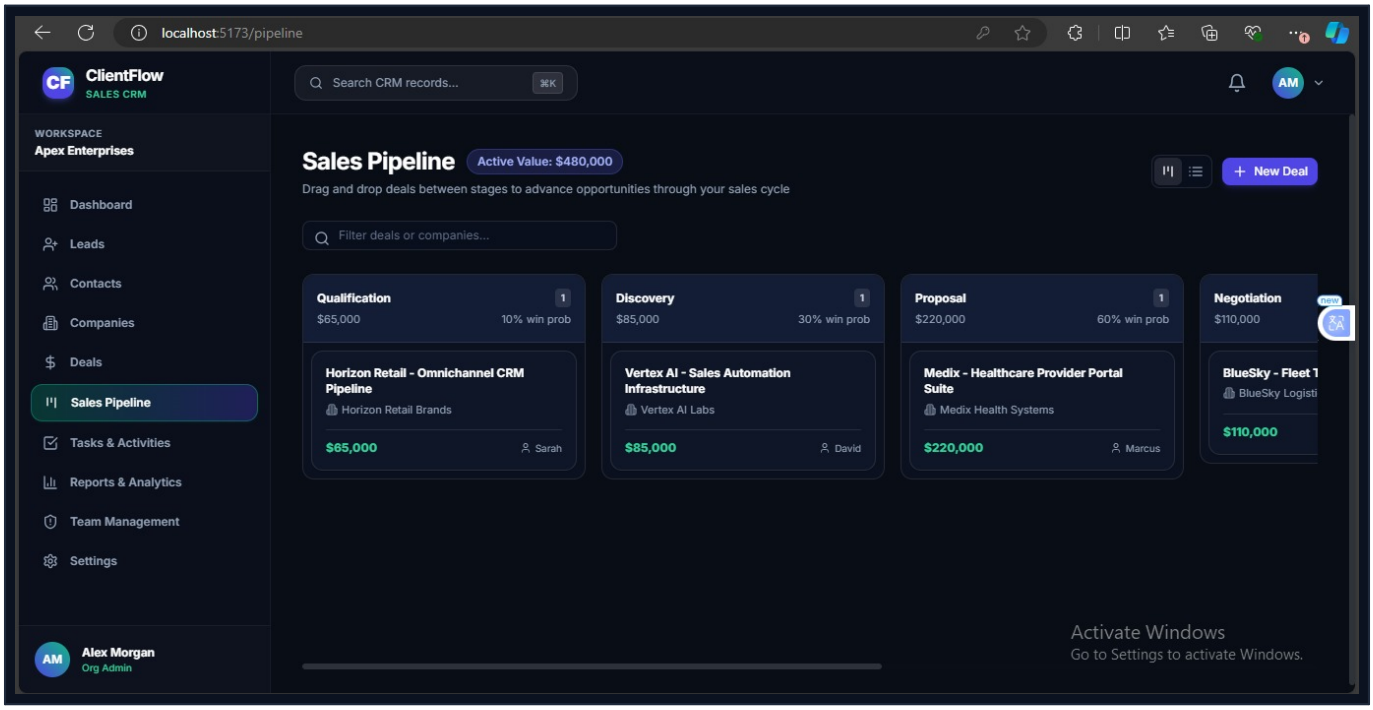


Figure 4.12 — Drag-and-drop Kanban board with stage totals, win probabilities and an Active Value badge ($480,000)

A view toggle switches between the Kanban board and the structured **Deals** table, which adds probability, owner and expected close date for reporting-style work and quick scanning.

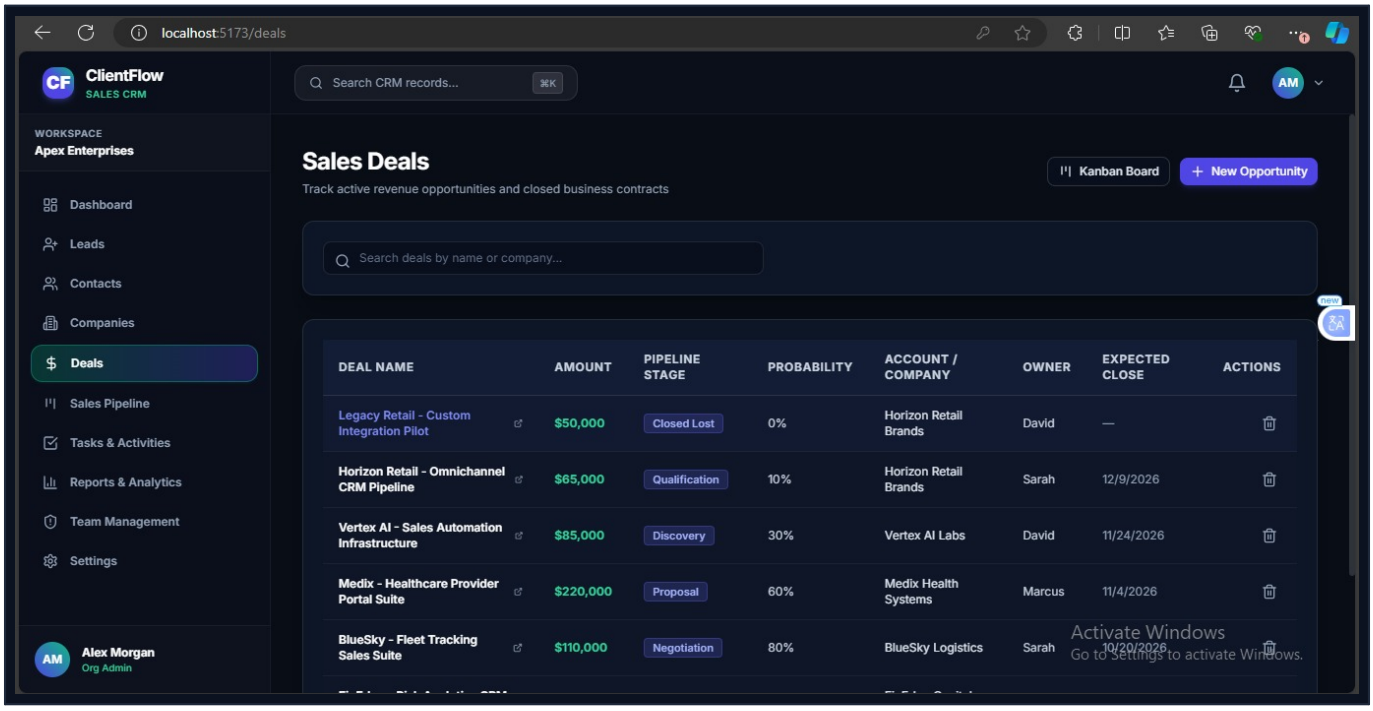


Figure 4.13 — Sales Deals table: amount, pipeline stage, probability, account, owner and expected close date

New opportunities are created from the **New Deal** button, which opens a compact form for the deal name, amount, pipeline stage and expected close date.

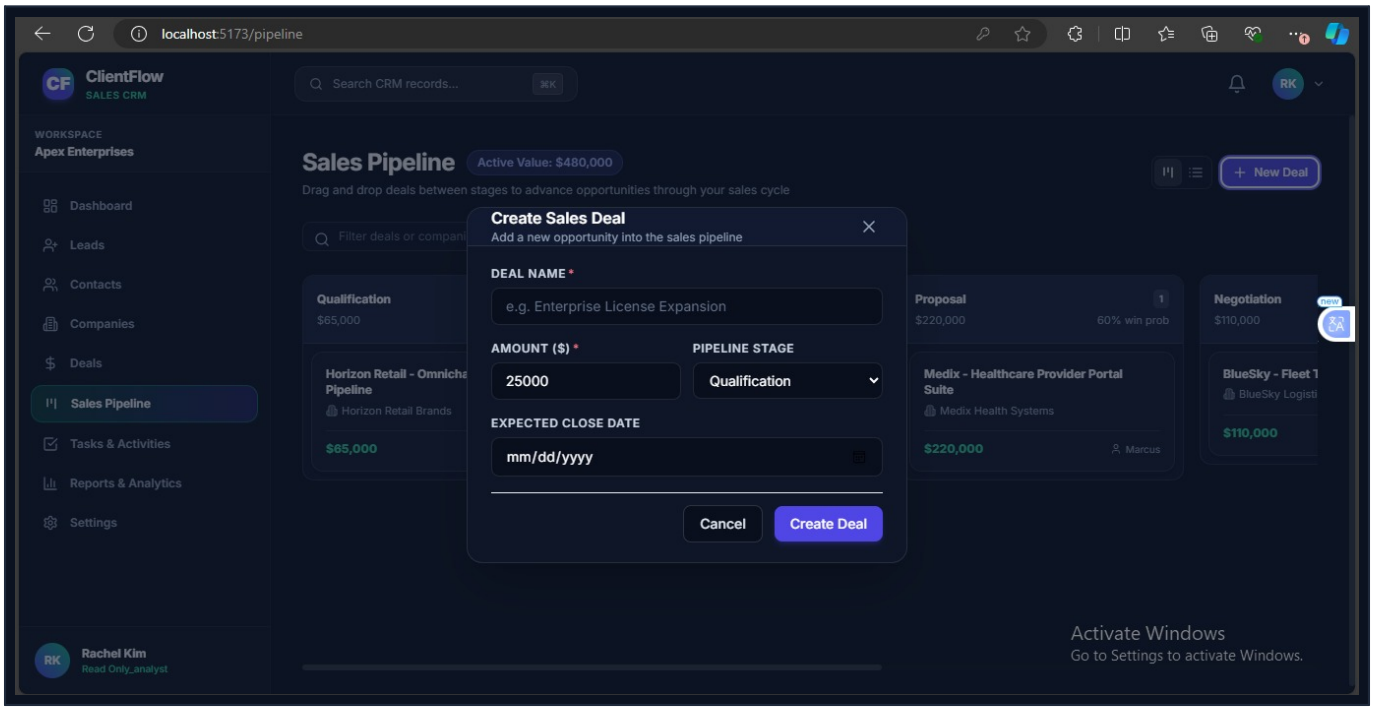


Figure 4.13b — Create Sales Deal modal opened from the pipeline

4.7 Tasks & Follow-Up Orchestration

Tasks keep calls, meetings and follow-ups on schedule. Four tabs (**All Tasks, Due Today, Overdue, Completed**) filter the list, overdue items are flagged in red with an alert icon, and priority badges (Urgent, High, Medium) show what to tackle first. A one-click checkbox completes a task and applies strike-through styling.

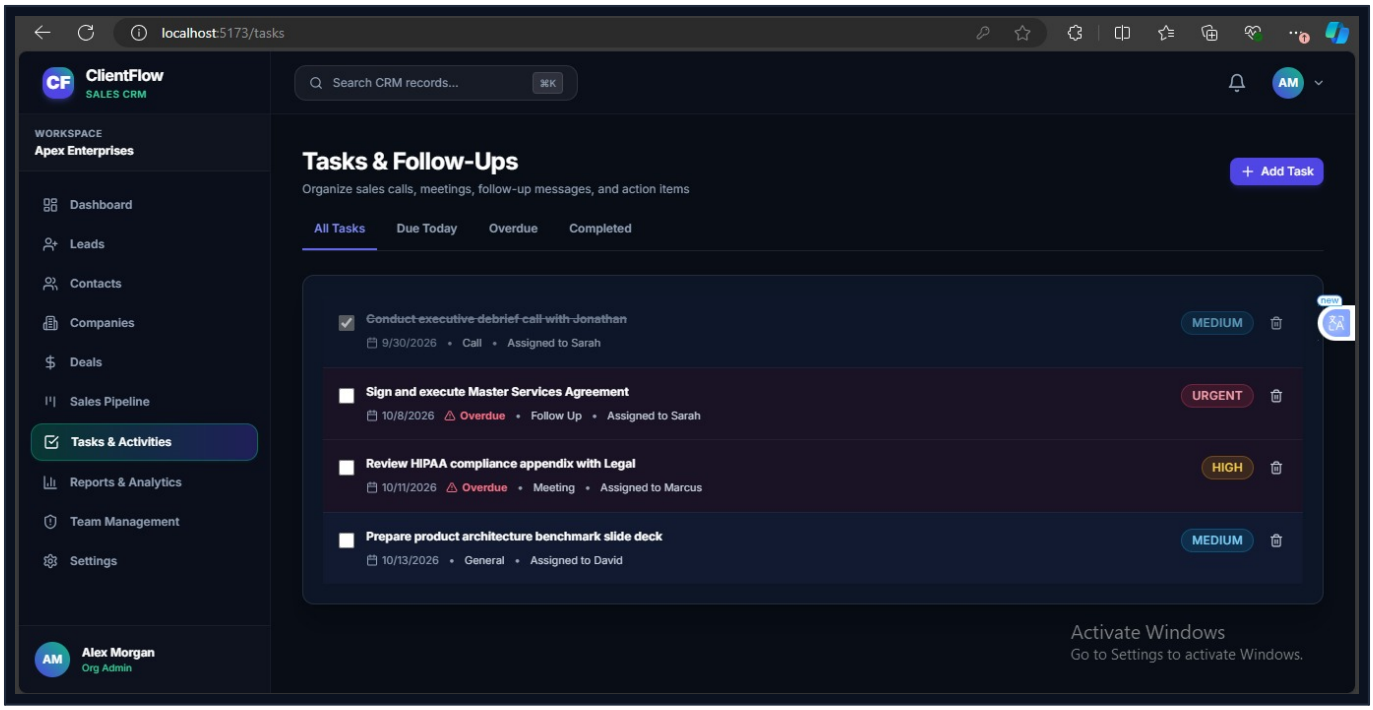


Figure 4.14 — Tasks & Follow-Ups with overdue warnings, priority badges and completion checkboxes

4.8 Reports, Analytics & CSV Export

Managers analyse sales velocity over a selectable period (*Last 7 Days, Last 30 Days, Current Quarter, Year to Date, All Time*). Summary cards report realized revenue, win rate, a weighted forecast and the lead conversion rate, while the representative productivity table ranks opportunities managed, deals won, win rate and realized revenue per person.

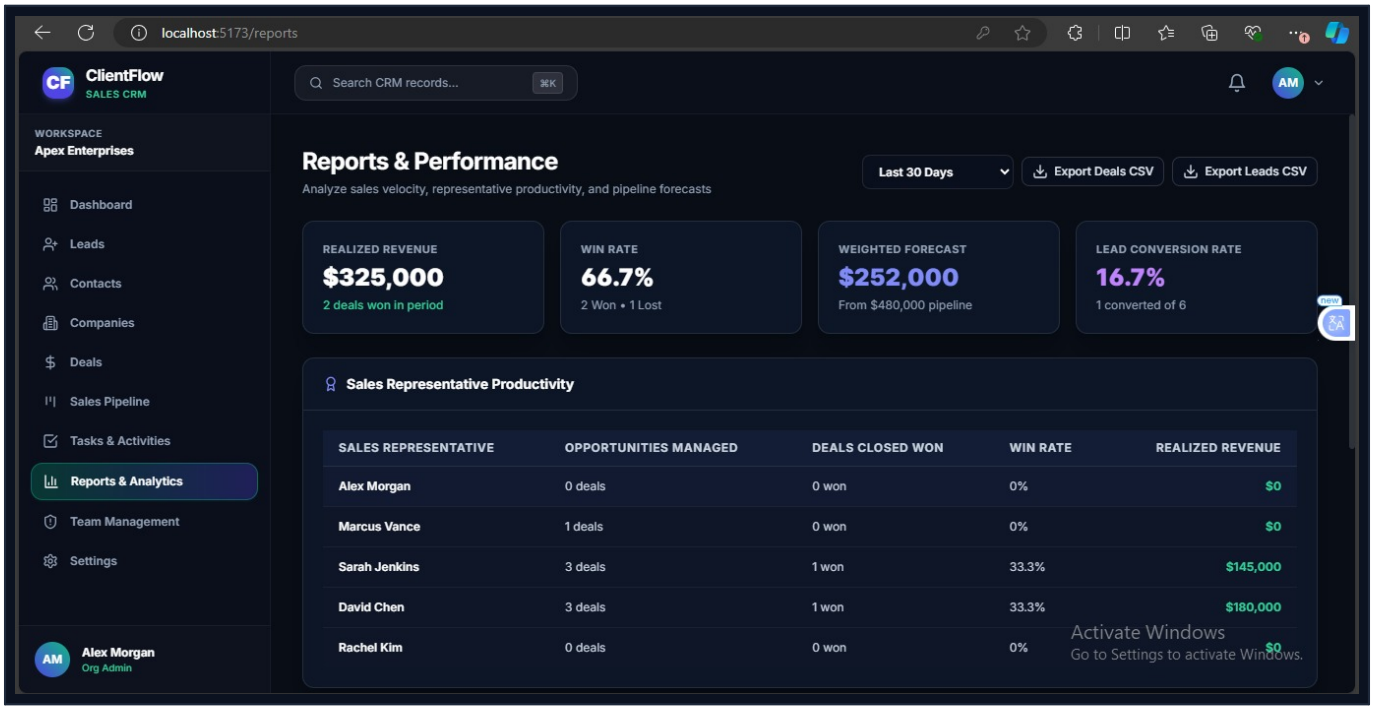


Figure 4.15 — Reports & Performance: KPI cards, period selector, CSV exports and representative productivity

Representative	Opportunities	Won	Win rate	Realized revenue
Sarah Jenkins	3 deals	1	33.3%	$145,000
David Chen	3 deals	1	33.3%	$180,000
Marcus Vance	1 deal	0	0%	$0
Alex Morgan / Rachel Kim	0 deals	0	0%	$0

Spreadsheet-safe exports

The **Export Deals CSV** and **Export Leads CSV** buttons download sanitised files. Any cell that begins with =, +, - or @ is prefixed with a single quote, which prevents formula injection when the file is opened in Excel or Google Sheets.

4.9 Team Management & Role-Based Access Control

Administrators manage the whole organization from one table: change a member's role from a dropdown, see each person's active leads and deals, invite new members, and activate or deactivate accounts instantly.

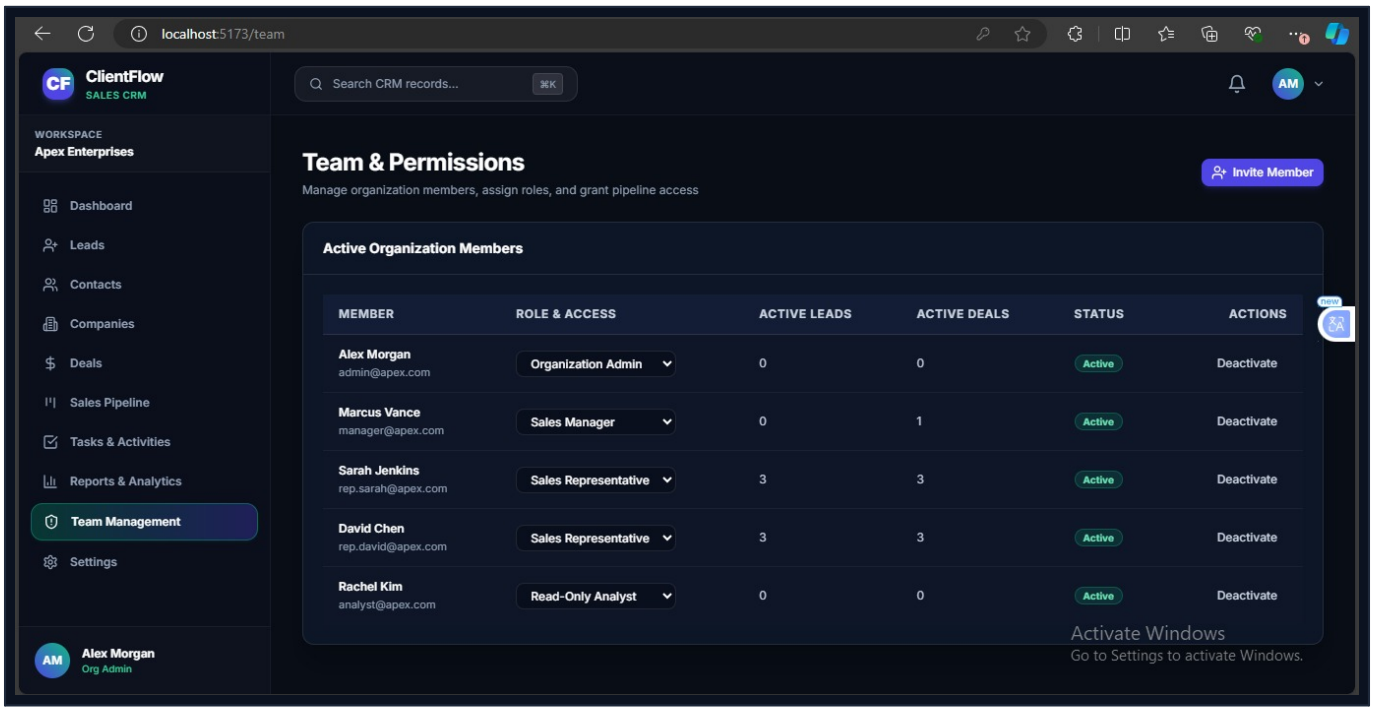
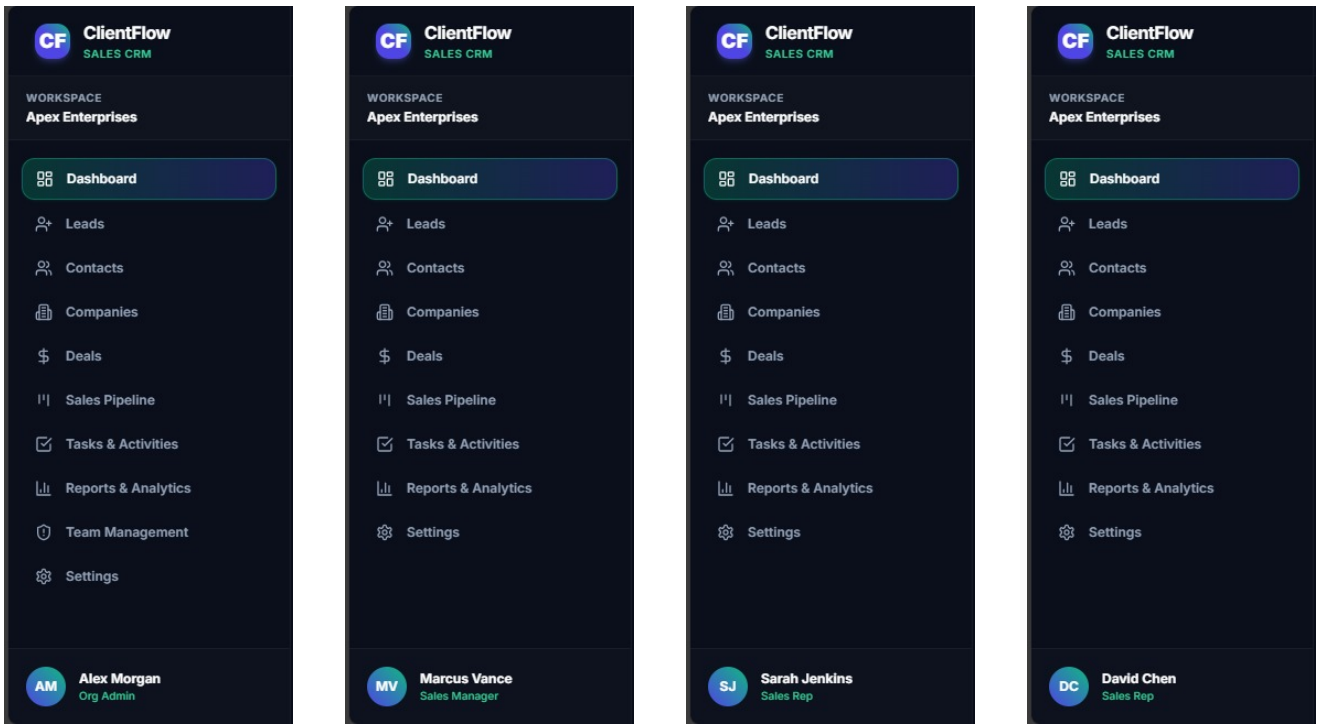


Figure 4.16 — Team & Permissions: role assignment, workload counts, status and Invite Member

Access control shows up in the interface itself. The sidebar is built from the signed-in user's role, so each person sees only the modules they are allowed to use, and the API enforces the same rules independently on the server.



Alex Morgan · Org Admin | **Marcus Vance · Sales Manager** | **Sarah Jenkins · Sales Rep** | **David Chen · Sales Rep**

Figure 4.17 — Role-aware navigation: the Org Admin sees every module, other roles see a reduced menu

Permission matrix

Capability	Org Admin	Sales Manager	Sales Rep	Read-Only Analyst
View dashboard, leads, contacts, companies, deals	✓	✓	✓	✓
Create and edit leads, contacts, deals and tasks	✓	✓	✓ (assigned)	–
Convert leads and move deals on the Kanban board	✓	✓	✓	–
Reports, analytics and CSV export	✓	✓	–	✓
Invite members and assign roles	✓	–	–	–
Workspace settings (name, currency, timezone)	✓	–	–	–

4.10 Workspace Settings

Administrators maintain the personal profile (name, email, role and workspace) and the organization configuration: the workspace name, the primary reporting currency and the system timezone used for due dates and reporting periods.

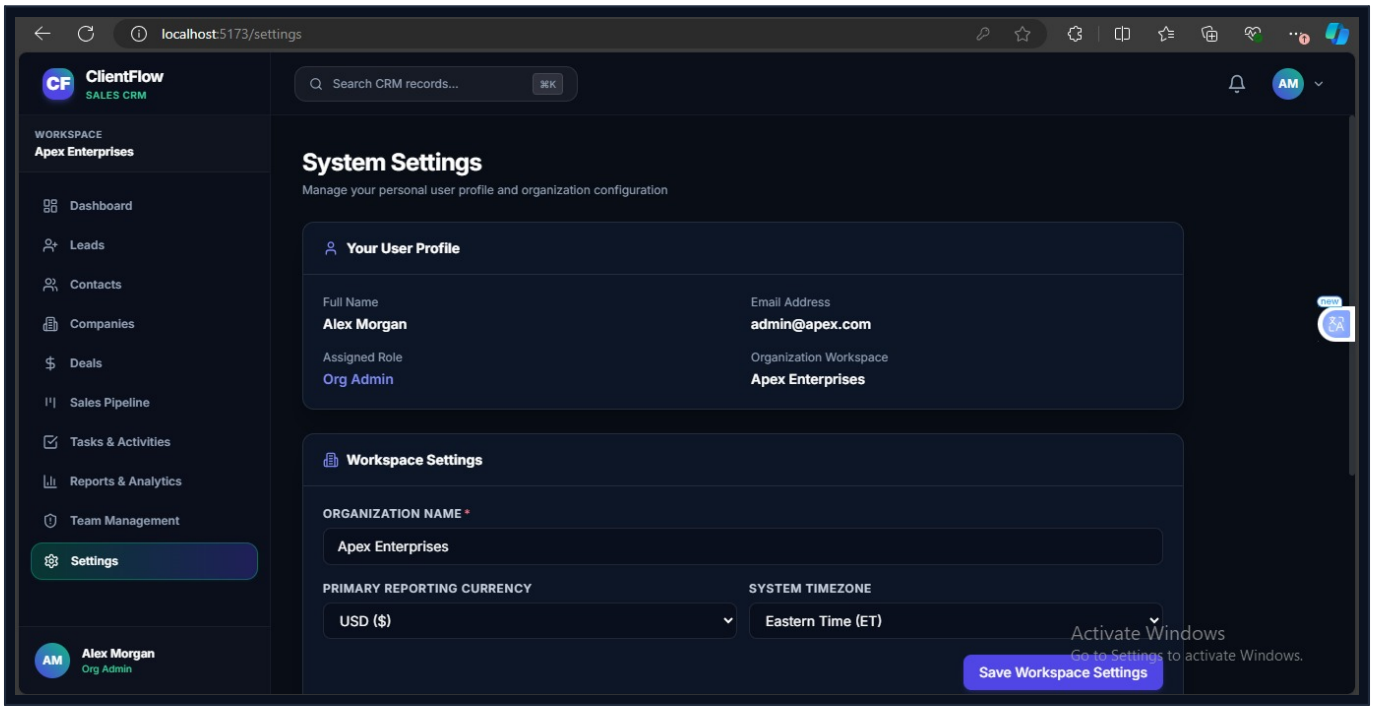


Figure 4.18 — System Settings: user profile card and workspace configuration form

5 Demo Environment

A pre-seeded organization so every role can be explored in seconds.

Running the seed script populates a demo organization, **Apex Enterprises**, with a sales pipeline, leads, companies, contacts, deals and tasks. Use any of the accounts below to sign in at `http://localhost:5173/login`. You can also register your own organization at any time on the `/register` page.

Role	Name	Email address	Password	Typical use
Organization Admin	Alex Morgan	`admin@apex.com`	`Password123!`	Full workspace access, invitations, settings
Sales Manager	Marcus Vance	`manager@apex.com`	`Password123!`	Pipeline oversight, team assignment
Sales Representative	Sarah Jenkins	`rep.sarah@apex.com`	`Password123!`	Leads, deals, tasks, notes
Sales Representative	David Chen	`rep.david@apex.com`	`Password123!`	Leads, deals, tasks, notes
Read-Only Analyst	Rachel Kim	`analyst@apex.com`	`Password123!`	Reports and analytics inspection

Demo dataset at a glance

$325,000
Won revenue (2 deals)

$480,000
Open pipeline (4 deals)

$252,000
Weighted forecast

66.7%
Win rate (2 won, 1 lost)

The six demo leads and the five demo companies (Horizon Retail Brands, Vertex AI Labs, Medix Health Systems, BlueSky Logistics and FinEdge Capital Partners) cover retail, AI, healthcare, logistics and financial services, giving each analytics view realistic variety. Because every figure is computed by SQL aggregation, creating or moving a record immediately changes the dashboard and report numbers.

6 API Endpoint Reference

Base URL: http://localhost:5000/api/v1 · Authentication: JWT bearer token

Method	Endpoint	Description	Auth	Minimum role
POST	`/auth/register`	Register a new organization and its initial admin	No	Public
POST	`/auth/login`	Authenticate a user and issue a JWT	No	Public
GET	`/auth/me`	Fetch the authenticated user profile	Yes	Any
GET	`/dashboard/stats`	Real-time KPI metrics and chart data	Yes	Any
GET	`/leads`	List leads (search and filters)	Yes	Any
POST	`/leads`	Create a new sales lead	Yes	Sales Rep
POST	`/leads/:id/convert`	One-click convert to Contact, Company and Deal	Yes	Sales Rep
GET	`/contacts`	List the customer contacts directory	Yes	Any
GET	`/companies`	List target enterprise accounts	Yes	Any
GET	`/deals`	List sales opportunities	Yes	Any
PATCH	`/deals/:id/stage`	Update deal stage (Kanban drag and drop)	Yes	Sales Rep
GET	`/pipeline`	Fetch the active pipeline and stage definitions	Yes	Any
GET	`/tasks`	List tasks filtered by status	Yes	Any
PATCH	`/tasks/:id/toggle`	Toggle task completion	Yes	Sales Rep
GET	`/reports/summary`	Sales analytics and rep productivity rankings	Yes	Sales Manager
GET	`/reports/export/:type`	Download sanitised CSV (deals or leads)	Yes	Sales Manager
GET	`/team/members`	List organization team members	Yes	Org Admin
PATCH	`/team/members/:id/role`	Update a member's permission role	Yes	Org Admin

Health check

A liveness endpoint is available at `GET /api/v1/health` on port 5000 and is the quickest way to confirm the API and database connection are up.

7 Security, Testing & Quality

Defence in depth from the database query to the CSV download.

Multi-tenant data isolation

Tenant scoping is enforced at the database-query level using authenticated JWT claims. Automated tests assert zero cross-tenant leakage.

Password security

Argon2id / bcrypt hashing with 12 salt rounds. Plain-text passwords are never logged or stored.

Formula-injection prevention

CSV values starting with =, +, - or @ are prefixed with a single quote so spreadsheets treat them as text.

Rate limiting

Brute-force protection on /auth/login and /auth/register using Express-Rate-Limit.

Input validation

Zod schemas validate API request bodies, queries and parameters as well as frontend forms; Prisma parameterises every query to prevent SQL injection.

HTTP hardening

Helmet security headers and a CORS policy restricted to the configured frontend origin.

Testing & quality assurance

17/17
API integration tests passing

0
Compiler or lint errors in the production build

The backend ships with an automated Supertest + Jest integration suite that exercises authentication, role enforcement, tenant isolation and the core CRM workflows. The frontend builds cleanly with Vite, and the whole monorepo passes a full TypeScript compilation check.

```
# Run the backend API integration test suite
npm run test --workspace=@clientflow/api

# Full TypeScript compilation check + frontend production build
npm run build
```

8 Getting Started & Local Setup

From clone to running application in four steps.

Prerequisites

- Node.js v18.x or v20.x (recommended)
- PostgreSQL running locally on port 5432, or started through Docker (`docker-compose.yml`)

Step 1 — Install dependencies

```
npm install
```

Step 2 — Configure environment variables

Create or verify the `.env` file in the project root. Generate long random values for the JWT secrets and never commit real credentials to version control.

```
DATABASE_URL="postgresql://<user>:<password>@localhost:5432/salesmanagement_db?schema=public"
PORT=5000
FRONTEND_URL=http://localhost:5173
JWT_ACCESS_SECRET=<long-random-string>
JWT_REFRESH_SECRET=<another-long-random-string>
NODE_ENV=development
```

Tip: if the PostgreSQL password contains special characters such as @ or #, percent-encode them inside the URL (@ becomes %40, # becomes %23).

Step 3 — Initialise the schema and seed demo data

```
# Push the Prisma schema to PostgreSQL
npm run db:push

# Seed the demo organization, pipeline, leads and deals
npm run db:seed
```

Step 4 — Start the development servers

```
npm run dev    # runs the API and the web app concurrently
```

Service	URL
Frontend web app	`http://localhost:5173`
Backend REST API	`http://localhost:5000`
API health check	`http://localhost:5000/api/v1/health`
Live documentation	`http://localhost:5173/doc.html`

License

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